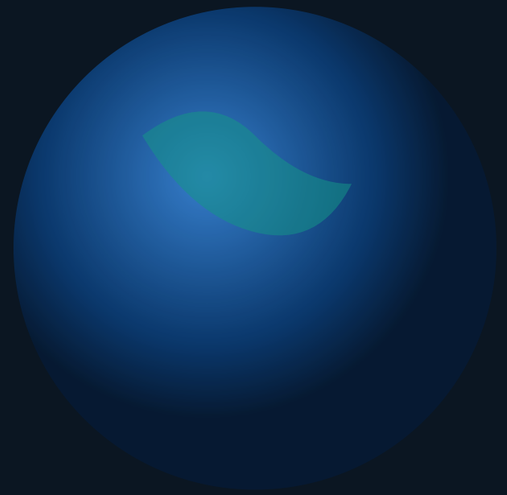
WEEK 1 · DAY 1

Colab Bootcamp + Your First Time Series

Today you open Google Colab, load real satellite-style data and draw your first time-series chart. 🛰️



NIGCOMSAT
AFRICAN ROOTED GLOBALLY POSITIONED

Publica Academy

TODAY'S MISSION · 2 HOURS

Three moves. That's all.



Open Colab

your Python lives in the browser



Load the data

3 years of Ka-band availability



Draw the story

find the rainy-season dip yourself

⏱️ 10 min hook · 🎤 30 min slides · 👩🏾‍💻 60 min hands-on · 🏁 20 min wrap-up

QUICK QUESTION

Did NEPA bring light yesterday? 💡

Here is one house, hour by hour. Yellow = light. Dark = “NEPA has taken light.”



🕐 A time

every hour, 00 → 23

⚡ A value

light on (1) or off (0)

= a time series

You have been collecting data all your life. “Up NEPA!” is a data point. 😄

PART 1

Colab Bootcamp 💻

Your free Python computer that lives inside Google Chrome.

WHAT IS COLAB?

Your laptop just shows the screen. Google's computer does the work.



Your laptop

only needs Chrome



Google's computer

already has Python, pandas, matplotlib, scikit-learn



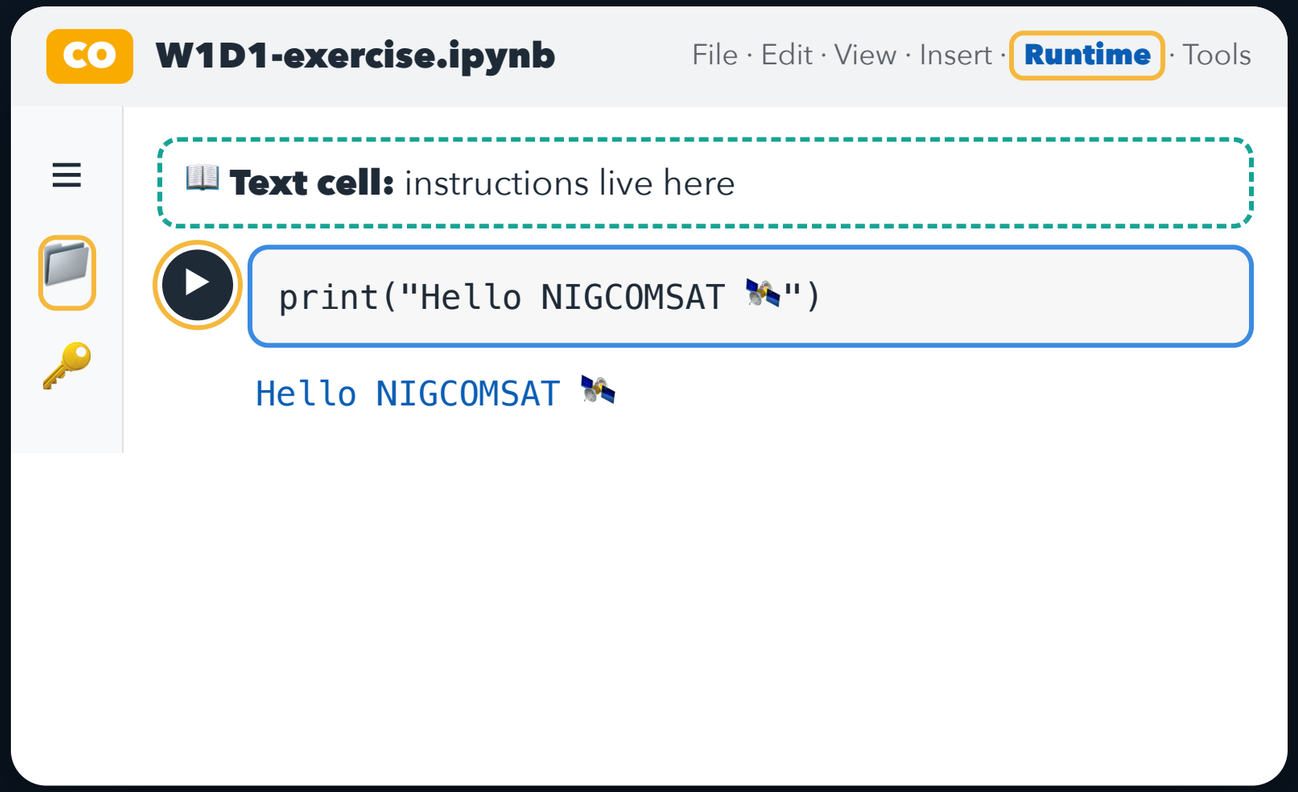
Your answers

tables + charts, back on your screen

Like a cyber café computer… **but free, and it never runs out of fuel.** ⛽😄

FIND YOUR WAY AROUND

The Colab screen in 5 parts



1. ▶ **Run button**: runs one cell. Shortcut: **Shift + Enter**
2. **Code cell** (blue box): Python goes here
3. **Text cell** (green box): reading, not running
4. 📁 **Files panel**: data you download or upload
5. **Runtime menu**: restart and “Run all”

COLAB'S 3 GOLDEN RULES

Follow these and Colab won't embarrass you



1. Save a copy first

File → **Save a copy in Drive**. Otherwise your work vanishes when you close the tab.



2. Top to bottom

Run cells **in order** with Shift + Enter. Skipping a cell is like skipping a step in a recipe.



3. Confused? Restart

Runtime → **Restart session and run all**. It fixes 80% of weird problems.

Rule 1 is the big one. Do it now, before anything else.

WHY DO I GET NAMEERROR?

Colab has a short memory 🧠

What Colab remembers right now

```
hours_in_year = 8760
```

```
ka = [1096 rows]
```

```
TIME_COLUMN = "timestamp"
```

🔄 Restart session…

…or leave Colab idle for too long. The memory is wiped. 🧹

```
NameError: name 'ka' is not defined
```

Translation: **“I don't remember. Run the cells above again.”**

THE AI BUTTON (GEMINI) IN COLAB

AI can help you. It must never see restricted data.



OK to paste

This course's training data (it is synthetic, made for teaching)
Your own code and error messages
“Explain this line to me”



Never paste

Real NIGCOMSAT telemetry, RF records, network logs
Customer or staff details
Anything your unit has not cleared

Not sure? **Describe the shape** (“a table of hourly voltages”), **don't paste the data.**

PART 2

What is a time series? ⏱️

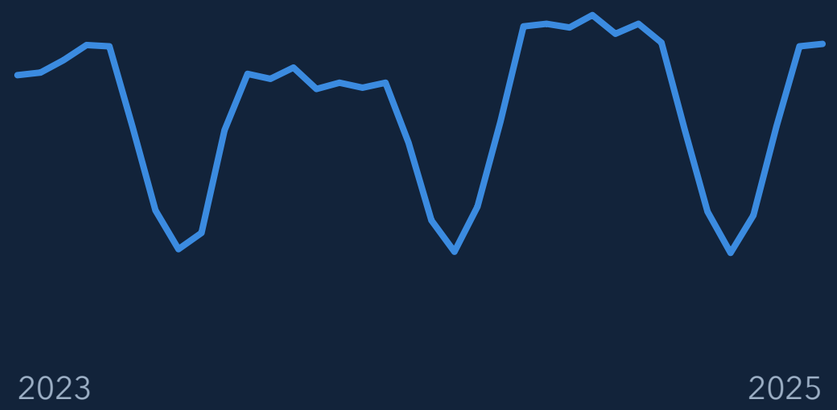
One idea. Once you get it, the whole course makes sense.

THE ONE BIG IDEA

A time series is a table where the ORDER of the rows carries meaning

✅ In time order

Ka-band availability, 36 months



You can see the yearly rainy-season dip.

🔀 Same numbers, shuffled

exactly the same 36 values



The story is gone. Order **is** information.

ANATOMY

Every time series has 3 things

🕐

A timestamp

when it happened

```
2025-08-01T21:00
```

🔢

A value

what was measured

```
battery_soc_pct = 77.3
```

🥁

A rhythm (cadence)

how often it is recorded

hourly **daily** **weekly** **monthly**

Like checking your phone: some people check **every minute**, some **once a day**. Same life, different cadence. 📱

MEET THE STAR OF THE COURSE

NigComSat-1R

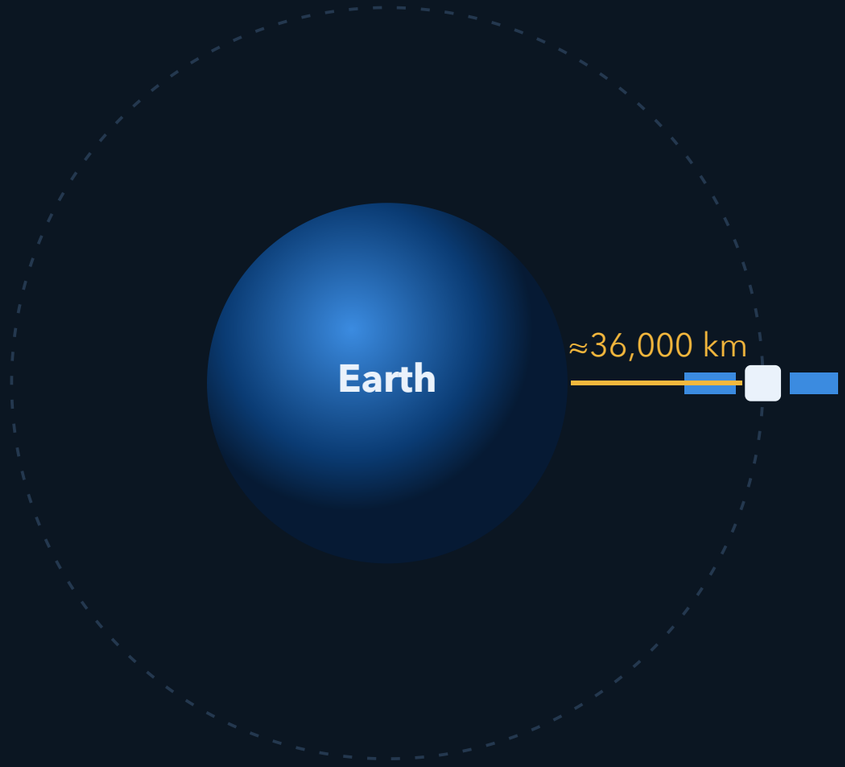
Nigeria's communications satellite, parked over the equator at **42.5° East**.

28
active transponders

36,000
km above the Earth (approx.)

C·Ku·Ka·L
four frequency bands

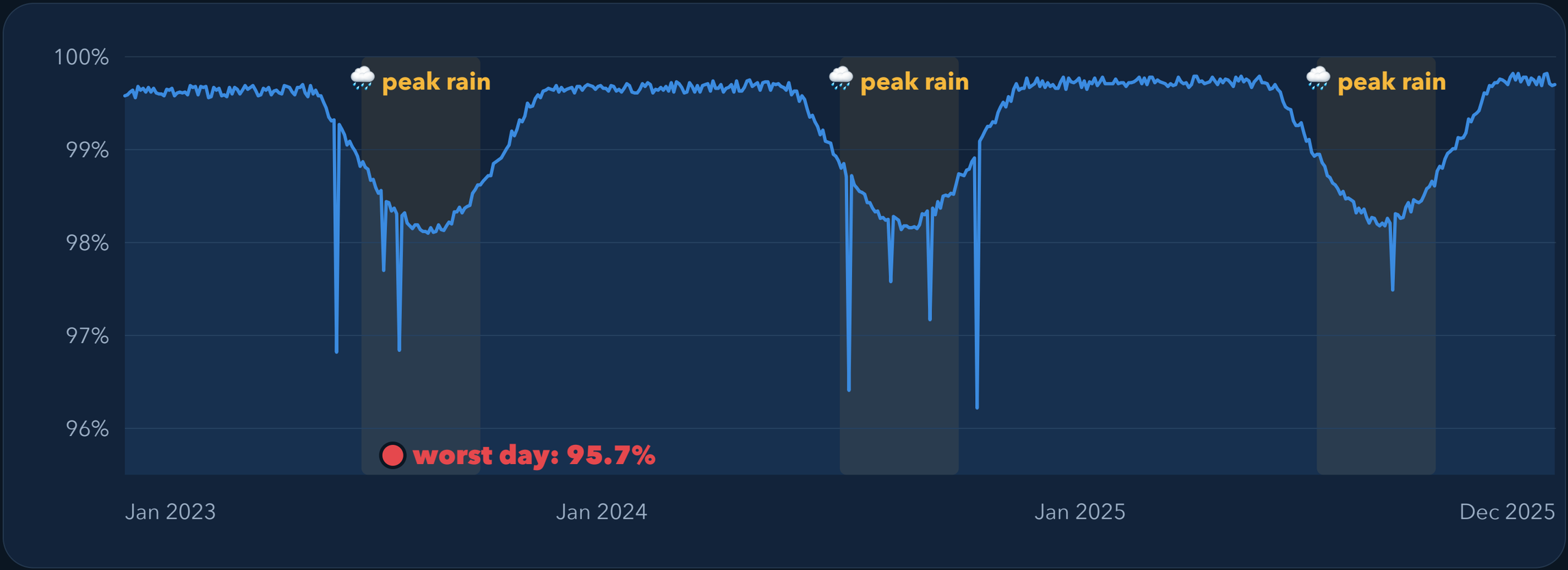
15 yrs
design life (launched 2011)



geostationary orbit: it stays over the same spot

TODAY'S DATASET · KA_BAND_AVAILABILITY_DAILY.CSV

3 years of the Ka-band spot beam over Nigeria



1,096 days

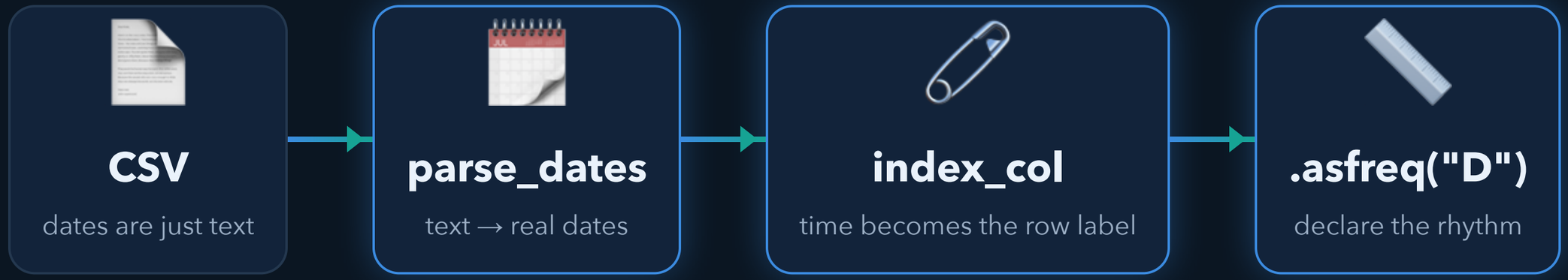
no gaps

dips every rainy season = rain fade

synthetic teaching data

TEACH PANDAS ABOUT TIME

The 3-line recipe you will use every day




```
ka = pd.read_csv(data_path("ka_band_availability_daily"),
                 parse_dates=["timestamp"],      # 1. text -> dates
                 index_col="timestamp")          # 2. time is the row label
ka = ka.asfreq("D")                              # 3. one row per day, gaps show up
```

WHY .ASFREQ()?

Missing days must show up, not hide

Export from a system: day 6, 7 and 8 never arrived. Can you see it?

1	2	3	4	5	9	10	11	12	13	14

11 rows. Looks fine. **It is lying.** 🙈

After `.asfreq("D")`: one slot for every day

1	2	3	4	5	NaN	NaN	NaN	9	10	11	12	13	14

14 rows. The 3 missing days appear as **NaN** (“Not a Number” = empty). Now you can count them. ✅

BEFORE YOU TRUST ANY NEW DATA

Inspect it like a tokunbo car 🚗🔍

Rows?

how many readings

Range?

first and last date

Gaps?

missing periods

Duplicates?

same time twice

Missing?

empty cells (NaN)

You check the engine, tyres and papers **before** you pay. Check these 5 **before** you analyse. 💸

TIME ZONES

Satellites speak UTC. Lagos speaks WAT.



21:00 UTC

satellite telemetry: eclipse time



22:00 WAT

Lagos / Abuja clock: UTC + 1, all year

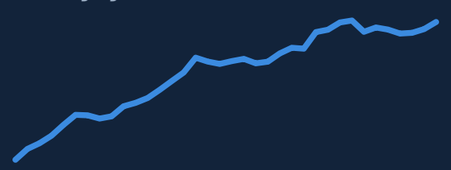
Rule: use ONE time zone before joining two tables. (Nigeria has no daylight saving 🙌)

SNEAK PEEK OF WEEK 2

Every series is made of 4 characters

Trend

The long direction. Like Lagos: more people every year.



transponder use, 2023→2025

Seasonality

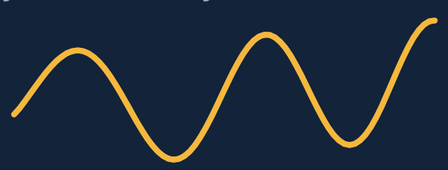
Repeats on a fixed calendar. Rainy season, every year.



Ka-band availability, monthly

Cycle

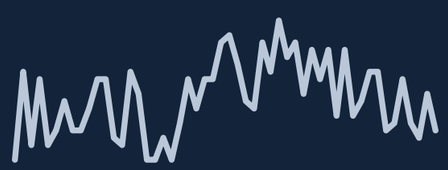
Ups and downs with NO fixed length. Boom years, slow years.



illustration

Noise

Random wobble. 38 tickets vs 41 tickets = same week.



NOC tickets per week, 2025

YOUR TURN · 60 MINUTES

Open the exercise notebook 👩🏾‍💻

1. Click **Open in Colab** on W1D1-exercise
2. **File → Save a copy in Drive** 💾
3. Run **STEP 0**, then go top to bottom
4. Replace every ____, run, get the ✅



12 checks to pass

Part A: Colab warm-up
Part B: your first time series
🌶️ Bonus: the pepper task

Keep the **rule book** open next to you.

TODAY IN 3 LINES

What you now know



Colab: save a copy, run top to bottom, restart when confused.



Time series: a table where **order matters**: timestamp + value + cadence.



pandas recipe: parse_dates → index_col → .asfreq() so gaps can't hide.

Next time 👉 **Charts That Don't Lie**: why the *average* hid a whole rain outage. 🌧️